

LAWTON FORT SILL SOUTH, OK, USA

WMO: 723575

Lat: 34.558N

Lon: 98.417W

Elev: 1069

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.4	20.3	3.6	6.9	22.4	8.9	9.0	27.9	30.5	39.1	27.0	40.1	8.0	0	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	104.1	73.0	101.2	73.2	98.9	73.5	78.0	92.5	77.0	91.5	76.0	90.5	10.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	132.5	83.6	73.1	127.9	82.2	72.3	124.4	81.7	42.2	92.0	41.1	91.4	40.1	90.3	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.2	23.2	20.1	DB	8.5	106.4	4.9	3.0	5.0	108.6	2.2	110.4	-0.5	112.1	-4.1	114.2	
			WB	7.4	79.8	4.6	1.5	4.1	80.9	1.4	81.8	-1.2	82.6	-4.5	83.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	40.7	44.8	54.0	61.9	71.2	80.2	84.4	84.1	76.0	63.7	51.6	41.8
	DBStd	17.54	9.14	10.20	9.74	8.23	7.48	5.21	4.52	5.03	6.84	8.70	9.15	8.83
	HDD50	967	309	201	69	11	0	0	0	0	0	8	89	280
	HDD65	3123	752	569	359	150	29	0	0	0	6	131	407	720
	CDD50	5698	22	54	193	368	658	906	1068	1056	780	431	137	25
	CDD65	2379	0	2	18	57	221	456	603	591	336	89	6	0
	CDH74	29889	12	47	238	702	2408	5482	8270	7872	3776	973	103	6
	CDH80	16225	0	7	54	216	1041	2930	5002	4726	1898	334	17	0
	WSAvg	9.3	8.9	9.9	10.8	10.9	10.1	9.9	8.3	8.0	8.3	9.0	9.0	8.7
	PrecAvg	29.1	1.1	1.3	2.2	2.5	4.4	3.8	2.0	2.5	3.3	2.8	1.8	1.5
Precipitation	PrecMax	41.7	3.2	4.6	6.8	6.2	16.3	10.3	8.4	9.4	7.5	11.1	6.1	5.8
	PrecMin	16.4	0.0	0.0	0.0	0.1	0.2	0.6	0.0	0.0	0.4	0.2	0.0	0.0
	PrecStd	7.4	0.9	1.0	1.7	1.7	3.2	2.5	1.8	2.2	2.0	2.7	1.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	79.7	86.1	91.5	99.2	104.0	107.4	107.9	103.6	92.4	82.4	73.1
		MCWB	54.9	59.2	61.3	65.6	70.6	72.7	72.2	71.8	70.7	69.1	63.9	56.0
	2%	DB	68.8	73.3	80.3	85.8	93.7	99.5	104.0	104.0	97.4	88.1	76.8	67.2
		MCWB	53.5	56.8	62.4	65.3	71.9	73.5	73.2	73.2	72.1	68.8	61.4	55.0
	5%	DB	63.4	68.0	75.6	81.5	90.0	96.3	101.1	101.2	93.5	83.5	72.3	62.8
		MCWB	50.6	54.3	61.0	65.1	70.9	73.5	73.7	73.2	72.0	66.3	59.4	52.0
	10%	DB	58.2	62.9	71.6	77.6	85.9	93.3	98.7	98.7	90.4	79.6	67.8	57.8
		MCWB	47.3	52.3	58.6	63.0	70.1	73.6	74.3	73.3	71.3	65.2	57.4	48.9
	0.4%	WB	61.6	64.2	68.3	72.9	76.9	79.5	79.6	78.6	77.4	74.6	68.1	63.7
		MCDB	68.7	71.8	78.2	82.5	90.0	93.4	95.7	93.0	90.2	84.0	76.0	68.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	56.4	60.6	65.8	70.1	75.2	77.8	78.0	77.4	75.5	72.6	64.5	58.3
		MCDB	63.9	68.5	75.2	79.3	87.1	91.4	93.0	92.5	88.0	81.9	71.9	63.8
	5%	WB	52.1	57.1	63.5	68.2	73.8	76.5	77.0	76.2	74.4	70.3	62.2	53.6
		MCDB	60.7	65.2	71.5	76.9	85.2	89.7	91.9	91.8	86.7	78.3	68.9	59.7
	10%	WB	48.1	52.7	60.5	66.2	72.1	75.2	75.9	75.2	73.2	68.1	59.4	49.2
		MCDB	58.0	62.3	69.2	74.3	82.6	88.1	91.2	90.8	85.4	76.1	66.9	57.5
Mean Daily Temperature Range	5% DB	MDBR	25.5	24.5	24.8	24.6	23.3	22.5	24.5	24.6	24.0	24.6	24.8	23.6
		MCDBR	36.5	34.5	31.6	30.8	29.2	27.2	29.4	29.2	28.3	30.0	31.5	32.8
		MCWBR	22.2	20.0	17.3	15.6	11.6	8.4	7.5	7.4	9.3	13.4	18.0	20.4
	5% WB	MCDBR	31.7	29.7	24.8	23.7	23.8	22.3	23.4	23.5	22.0	21.1	25.1	25.7
		MCWBR	21.2	20.4	15.1	14.2	11.5	9.2	7.7	7.4	8.1	11.2	16.7	19.1
Clear-Sky Solar Irradiance	taub		0.291	0.300	0.337	0.372	0.408	0.428	0.424	0.419	0.387	0.349	0.314	0.294
	taud		2.518	2.507	2.421	2.349	2.318	2.301	2.347	2.304	2.370	2.462	2.497	2.508
	Ebn at Noon		293	302	297	290	279	272	272	272	277	280	282	285
	Edh at Noon		26	29	35	39	41	42	40	41	36	30	26	25
All-Sky Solar Radiation	RadAvg		937	1126	1495	1829	1964	2178	2191	2006	1667	1280	988	796
	RadStd		95	160	95	147	209	189	130	116	135	178	125	88

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page